

Akhil Manoj

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EDUCATION

New York University, Tandon School of Engineering

New York City, U.S.A

Master of Science, Computer Engineering

Sept 2024 – May 2026

- GPA: 4.0/4.0
- Course Assistant: CSYG6033 Design and Analysis of Algorithms I

EXPERIENCE

Software Development Engineer Intern

May 2025 – Aug 2025

Amazon Web Services — Provable Security And Automation

New York, U.S.A

- **Built a CLI tool to containerize Model Context Protocol servers** with minimal-permission configuration, enabling secure sandboxing and reducing integration risks.
- **Developed an intelligent retry system** that monitored container failures, parsed logs for missing dependencies, and dynamically injected temporary resources to enable re-execution.
- **Added multi-language build support** for Python and Node.js with auto-detection, Git integration, Docker image creation, and streamlined CLI workflows.

Software Development Engineer

June 2022 – June 2024

Gupshup Pvt Ltd

Bengaluru, India

- **Led the development of the Ad-creation feature** within the Gupshup platform, enabling the creation of multimedia Facebook ads directly from Gupshup, enhancing the customer experience.
- **Designed and implemented a high-efficiency service** for fetching real-time ad insights, incorporating multithreaded batch requests, rate limiting, and notifications to improve data accuracy and timeliness for **25 business clients managing over 2,000 Facebook ads**.
- Cut deployment time by **55%** through custom Docker images, improving the application deployment process.

AR/VR Developer Intern

June 2019 – July 2019

Schneider Electric

Bengaluru, India

- **Contributed to an augmented reality application for Microsoft HoloLens**, enabling the identification of sensors via QR-codes and displaying real-time details on the HoloLens screen.
- **Developed a calibration tool** to ensure accurate alignment of virtual overlays with sensors, enhancing the precision of augmented reality interactions for users.
- **Integrated APIs to fetch real-time sensor data**, enabling users to view up-to-date metrics on the HoloLens interface.

PROJECTS

Dining-Concierge Chatbot with AWS | *Lambda, OpenSearch, Gateway, DynamoDB*

Oct 2025 – Nov 2025

- **Built a serverless Dining Concierge chatbot** using AWS Lex, API Gateway, Lambda, SQS, SES, DynamoDB, and Opensearch to deliver restaurant suggestions through natural conversations
- **Implemented the full backend pipeline**, including Lex intents with Lambda hooks, a queue-based recommendations worker, and a dataset of 1,000+ Manhattan restaurants scraped from the Yelp API
- **Developed and hosted the web frontend on S3 and connected it to secure API Gateway endpoints**, letting users chat with the bot and receive automated restaurant suggestions by email

Organic Playlist Recommendation System | *Grafana, Prometheus, Airflow, GPUs*

Jan 2025 – Apr 2025

- **Designed and implemented a machine learning system** for organic playlist recommendations to be integrated in music streaming services
- **Trained large-scale KNN and Bayesian Personalized Ranking (BPR) models** on NVIDIA A100 GPUs, optimizing retrieval and ranking for millions of user-playlist pairs
- **Deployed the full ML pipeline using Kubernetes**, orchestrating Airflow jobs, MLflow experiments, FastAPI services, and Metabase dashboards in production-like environments

TECHNICAL SKILLS

Programming Languages,: Java, C/C++, Javascript, Python, Matlab

Tools, Frameworks and Core skills: REST APIs, Git, SpringBoot, SpringCloud, MySQL, Postgres, Spark, Docker, Kubernetes, Redis, RabbitMQ, React, Redux, TailwindCSS, AWS, Microservices, Multithreading, Design Patterns